ZACKARY CROSLEY

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EDUCATION

MASTER'S OF SCIENCE

Computer Science · Arizona State University · GPA 3.80 · Aug 2019

BACHELOR'S OF SCIENCE

Computer Science · Arizona State University · GPA 3.76 (Magna Cum Laude) · Dec 2017

EXPERIENCE

SOFTWARE ENGINEER II

Engineering · Chainguard · Jun 2025 – Present

- Implemented secure Azure account OIDC impersonation, including backend services and terraform infrastructure, to enable safe Azure account sign-in and OCI repository sync.
- Updated CLI tool chainctl to manage Azure account connections and Azure Container Registry syncing.
- Created data pipeline build and publish OpenVEX, OSV, and SecDB security feeds to communicate vulnerability status in Chainguard library and container products to scanning partners and customers.
- Contributed to Grype security scanner to adopt Chainguard OpenVEX and OSV feeds to accurately assess Chainguard artifacts, ensuring an accurate open-source scanner for customer security posture.
- Migrated Chainguard Advisories to a new V2 system with increased data fidelity and faster datastore. Validated migrated data, fixed data problems, and communicated changes to customers.
- Developed APIs for V2 Advisories, implementing business logic for internal teams to manage advisories and for external customers to efficiently query nearly a million security advisories.
- Developed TUI for managing advisories using Bubble Tea for internal teams to easily create, update, and collaborate on Chainguard Advisories, improving internal vulnerability management and tracking.
- Created system to modify Chainguard Helm Charts to automatically publish as AWS EKS Addons in AWS Marketplace, providing customers with updated, FIPS-compliant, and secured chart addons.
- Created event-driven scanning infrastructure to build scanner SQLite database, clone to regional buckets, and update running scanner services to ensure updated vulnerability assessments.

SOFTWARE ENGINEER II

Engineering · Censys Inc · Jun 2022 – May 2025

- Developed APIs, pipelines, and architecture for Exposure Management and Platform products.
- Designed and developed a pipeline for CVE Risks pulling NVD and host data over gRPC, generating risk definition, posting annotations over Pub/Sub, and loading into Google Spanner.
- Built service to poll customer networks for critical risks and update Postgres risk datastore, improving visibility into customer attack surfaces from full day refresh to minutes.
- Created gRPC Gateway to interface between Censys and Workato to manage third-party integrations, adding dozens of new integrations and reducing development time for new integrations.
- Developed a more efficient procedure for fingerprinting vulnerabilities in customer infrastructure. Methodology facilitated the rapid addition of dozens of new risk fingerprints.
- Led development of a service that managed credits for users and organizations. A scheduled job handled credit renewal and generated daily usage reports to support product, marketing, and sales.
- Contributed to open source project Formance Ledger adding business features for credit system.
- Developed TUI to interact with Censys data, simplifying and increasing API adoption for customers.

OPERATIONS RESEARCH ANALYST GS-12

The Research and Analysis Center · Army Futures Command · May 2017 – May 2022

- Served as developer on COMBATXXI, a stochastic (Monte Carlo), discrete event combat simulation.
- Developed a navigation suite in Python saving hundreds of man hours in development per scenario.
  - Generated graph data structure from terrain, storing contextual data (trafficability, node or edge type) for knowledge representation, search algorithms (A*, Dijkstra, ACO), and path manipulation.
  - Optimized graph via functional programming approach with memoization and lazy evaluation.
- Developed Utility AI framework in Java to define utility decisions for autonomous agent behaviors.
- Developed Java terrain reasoning methodology as sequential discrete sampling and utility evaluation.

STUDENT RESEARCHER

Security Engineering for Future Computing Lab · Arizona State University · Jul 2017 – Aug 2019

- Worked with Professor Adam Doupé and SEFCOM lab to generalize inductive programming technique presented by Sumit Gulwani (Microsoft). Methodology uses Directed Acyclic Graph intersection to isolate operations that could result in individual characters of output.
- Created novel technique to synthesize exploits in Capture the Flag (CTF) by sniffing traffic, categorizing exploits, learning logic using inductive programming, and reflecting exploits at other players.
- Documented research in thesis ARCHES and defense at Arizona State University.

PUBLICATIONS

AUTOMATED REFLECTION OF CTF HOSTILE EXPLOITS (ARCHES)

Master's Thesis · Arizona State University · 2019

PROFICIENCIES

Golang	Postgres	DevOps	OCI/Docker	Pub/Sub
Python	Neo4j	CI/CD	Terraform	Linux
Rust	Protobuf	OAuth	Git	Bash
Java	gRPC	OTel	GCP	Vim
Spanner	Redis	Kubernetes	AWS	Nix

HONORS AND ACTIVITIES

Department of the Army Civilian Service Achievement Medal · 2022
Operations Research Military Applications Course Honor Graduate · 2020
Science, Mathematics, and Research for Transformation Scholarship · 2016 · 2018

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